

Report: Integration of External Datasets for Waste Segmentation

Overview

In response to Client's request to explore external datasets for integration into the existing segmentation dataset of waste images, we conducted an in-depth review of available datasets. Given the high priority of this task, we carefully assessed multiple datasets focused on waste images to identify those that would meet your specific requirements.

Out of a comprehensive list of 23 waste image datasets, only four offered segmentation-type data specifically for waste. However, upon closer examination, only the TACO (Trash Annotations in Context) dataset aligns with the needs and has been successfully integrated into our current system. Unfortunately, the other segmentation datasets had significant limitations that rendered them unsuitable for our purposes, as detailed below.

If there is a continued emphasis on adding more external data, the only viable option would involve using datasets containing generic images of items (non-waste images). While these could potentially supplement certain segmentation tasks, they introduce unique challenges:

- Context Mismatch: Generic item images lack the specific context of waste, which may reduce model accuracy for waste-specific tasks.
- Increased Processing Overheads: Integrating such datasets will involve substantial preprocessing and may necessitate model retraining to mitigate the noise introduced by unrelated images. [Example](#)

The following sections provide a summary of our findings regarding each dataset considered.

Dataset Evaluation Summary

1. TACO (Trash Annotations in Context) - The only viable dataset that met our criteria for waste segmentation.
 - Successfully transformed and loaded into the computing cluster. ELT pipeline ready.
2. TrashCan 1.0 - This dataset focuses on underwater waste images, which introduces a distinct environmental context that does not align with our ground-based waste segmentation needs.
 - Not suitable.
3. UAVWaste - This dataset captures waste imagery from aerial perspectives via drones, leading to angles and resolutions that differ significantly from ground-level images.
 - Not suitable.

4. MJU-Waste v1.0 - Although this dataset theoretically supports segmentation, the lack of published labels (as per [MJU-Waste GitHub Issue #2](#)) makes it unusable for our purposes.
- Not suitable.

Given these limitations, we recommend proceeding with the TACO dataset as the primary external source for segmentation tasks. Should further segmentation data become essential, we will need to prepare for the challenges associated with adapting generic image datasets to meet waste segmentation standards.

Appendix: Evaluated Waste Datasets

Dataset Name	Type	Note
[TrashCan 1.0](https://conservancy.umn.edu/handle/11299/214865)	Instance-Segmentation	Underwater images; unsuitable for ground-based segmentation
[UAVVaste](https://github.com/PUTvision/UAVVaste)	Segmentation	Drone-based aerial imagery; incompatible perspective for ground segmentation
[MJU-Waste v1.0](https://github.com/realwecan/mju-waste/)	Segmentation	Labels not available; unusable for segmentation
[TACO](http://tacodataset.org/stats)	Segmentation	Integrated successfully; meets client requirements